

Problem 1.1.

- (a) Let Y be the plane curve $y = x^2$ (i.e. Y is the zero set of the polynomial $f = y - x^2$). Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k .
- (b) Let Z be the plane curve $xy = 1$. Show that $A(Z)$ is not isomorphic to a polynomial ring in one variable over k .
- (c) Let f be any irreducible quadratic polynomial in $k[x, y]$, and let W be the conic defined by f . Show that $A(W)$ is isomorphic to $A(Y)$ or $A(Z)$. Which one is it when?

Solution.

- (a) $A(Y) = k[x, y] / \langle x^2 - y \rangle \cong k[x]$
- (b) $A(Y) \cong k[x, y] / \langle xy - 1 \rangle \cong k[x, x^{-1}]$
- (c) Take general equation and analyse cases.
Assuming $\text{char}(k) \neq 2$.

$$\begin{aligned}
 f(x, y) &= ax^2 + bxy + cy^2 + dx + ey + f \\
 &= a\left(x^2 + \frac{b}{2a}y\right)^2 + cy^2 + dx + ey + f \\
 &=
 \end{aligned}$$

Problem 1.2. The Twisted Cubic Curve.

Let $Y \subseteq \mathbb{A}^3$ be the set

$$Y = \{(t, t^2, t^3) \mid t \in k\}.$$

Show that Y is an affine variety of dimension 1. Find generators for the ideal $I(Y)$. Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k . We say that Y is given by the *parametric representation*

$$x = t, \quad y = t^2, \quad z = t^3.$$

Solution.

Since k is a field, if $t = 0$ then $t^2 = t^3 = 0$, and the only ideals k has are 0 and k . So $\dim(Y) = \dim(A/I(Y)) =$

Problem 1.3. Let Y be the algebraic set in $\mathbb{A}^3$ defined by the two polynomials

$$x^2 - yz \quad \text{and} \quad xz - x.$$

Show that Y is a union of three irreducible components. Describe them and find their prime ideals.

Solution. Note that $xz - x = x(z - 1)$ and $x^2 - yz$ is irreducible. So $Y = \mathbb{V}(x^2 - yz, x, z - 1) = \mathbb{V}(x^2 - yz) \cup \mathbb{V}(x) \cup \mathbb{V}(z - 1)$. Their prime ideals are the ones they generate because they are irreducible.

Problem 1.4. If we identify $\mathbb{A}^2$ with $\mathbb{A}^1 \times \mathbb{A}^1$ in the natural way, show that the Zariski topology on $\mathbb{A}^2$ is not the product topology of the Zariski topologies on the two copies of $\mathbb{A}^1$.

Solution.

Problem 1.5. Show that a k -algebra B is isomorphic to the affine coordinate ring of some algebraic set in $\mathbb{A}^n$, for some n , if and only if B is a finitely generated k -algebra with no nilpotent elements.

Problem 1.6. Any nonempty open subset of an irreducible topological space is dense and irreducible. If Y is a subset of a topological space X , which is irreducible in its induced topology, then the closure $\overline{Y}$ is also irreducible.

Problem 1.7.

- (a) Show that the following conditions are equivalent for a topological space X :
 - (i) X is noetherian;
 - (ii) every nonempty family of closed subsets has a minimal element;
 - (iii) X satisfies the ascending chain condition for open subsets;
 - (iv) every nonempty family of open subsets has a maximal element.
- (b) A noetherian topological space is *quasi-compact*, i.e., every open cover has a finite subcover.
- (c) Any subset of a noetherian topological space is noetherian in its induced topology.

- (d) A noetherian space which is also Hausdorff must be a finite set with the discrete topology.

Problem 1.8. Let Y be an affine variety of dimension r in $\mathbb{A}^n$. Let H be a hypersurface in $\mathbb{A}^n$, and assume that $Y \not\subseteq H$. Then every irreducible component of $Y \cap H$ has dimension $r - 1$. (See (7.1) for a generalization.)

Problem 1.9. Let $\mathfrak{a} \subseteq A = k[x_1, \dots, x_n]$ be an ideal which can be generated by r elements. Then every irreducible component of $Z(\mathfrak{a})$ has dimension at least $n - r$.

Problem 1.10.

- (a) If Y is any subset of a topological space X , then $\dim Y \leq \dim X$.
 (b) If X is a topological space which is covered by a family of open subsets $\{U_i\}$, then

$$\dim X = \sup \dim U_i.$$

- (c) Give an example of a topological space X and a dense open subset U with

$$\dim U < \dim X.$$

- (d) If Y is a closed subset of an irreducible finite-dimensional topological space X , and if

$$\dim Y = \dim X,$$

then $Y = X$.

- (e) Give an example of a noetherian topological space of infinite dimension.

Problem 1.11. Let $Y \subseteq \mathbb{A}^3$ be the curve given parametrically by

$$x = t^3, \quad y = t^4, \quad z = t^5.$$

Show that $I(Y)$ is a prime ideal of height 2 in $k[x, y, z]$ which cannot be generated by 2 elements. We say Y is *not a local complete intersection* (cf. Ex. 2.17).

Problem 1.12. Give an example of an irreducible polynomial $f \in \mathbb{R}[x, y]$ whose zero set

$$Z(f) \subseteq \mathbb{A}_{\mathbb{R}}^2$$

is not irreducible (cf. 1.4.2).